IV Infusions Practice 4

Module 3 - Intravenous Infusions · Daily practice sheet · No date

Name _____ Date _____

The therapeutic plasma level of a drug is 4 to 15 mg/L. You would like to immediately attain a steady-state concentration level of 10 mg/L and maintain this concentration over 12 hours. The drug has a half-life of 6 hours and renal and metabolic clearances of 5 and 6.55 L/hr, respectively (total body clearance = renal clearance + metabolic clearance). The patient's body weight is 65 kg.

a Determine an IV bolus loading dose and infusion rate to achieve the desired steady-state plasma level of 10 mg/L. (Answer here: the infusion rate.)

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b Calculate the volume and concentration of the infusion solution if the rate of input of the solution is 1 mL/min. (Answer here: the volume.)

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c What is the apparent volume of distribution?

d If there were no loading dose, how much time would be required to reach the 5 mg/L level?

e If there were no loading dose, how much time would be required to reach 80% of the steady-state plasma concentration?

f Calculate the concentration of drug in the plasma at 1, 6 and 12 hours following the cessation of the infusion at the steady-state level of 10 mg/L. (Answer here: 1 hour after.)

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f Calculate the concentration of drug in the plasma at 1, 6 and 12 hours following the cessation of the infusion at the steady-state level of 10 mg/L. (Answer here: 12 hours after.)

g If the patient develops acute renal failure and the renal clearance is reduced to 2 L/hr, what IV bolus loading dose and infusion rate would you recommend to achieve the steady-state level of 10 mg/L? Assume metabolic clearance and the apparent volume of distribution remain constant. (Answer here: the infusion rate.)

g If the patient develops acute renal failure and the renal clearance is reduced to 2 L/hr, what IV bolus loading dose and infusion rate would you recommend to achieve the steady-state level of 10 mg/L? Assume metabolic clearance and the apparent volume of distribution remain constant. (Answer here: the IV bolus loading dose.)

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

a

Her key

Answer: 115.5 mg/hr

WHAT THE STEM GIVES

window	4 to 15 mg/L	*not needed: the target 10 mg/L is stated*
C_{ss}	10 mg/L	the target, times Cl_T
duration	12 hours	*not needed*
t½	6 hours	*not needed for the rate*
Cl_R	5	in L/hr, part of Cl_T
CIM	6.55 L/hr	$Cl_T = 5 + 6.55 = 11.55$ L/hr
weight	65 kg	*not needed*

THE WORKING, ONE LINE AT A TIME

ALGEBRA

$$Cl_T = Cl_R + CIM = (5 + 6.55)\ \text{L/hr} = 11.55\ \text{L/hr}$$

The stem states that total body clearance is renal plus metabolic clearance; both are in L/hr, so they add.

SETUP

$$R = C_{ss} \times Cl_T$$

At steady state the rate in equals the rate out, $Cl_T \times C_{ss}$, so the rate that holds 10 mg/L is the target times clearance.

ALGEBRA

$$R = (10\ \text{mg/L})(11.55\ \text{L/hr}) = 115.5\ \text{mg/hr}$$

mg/L × L/hr: the litres cancel and mg/hr is left, the amount infused each hour.

> **DOES THE NUMBER MAKE SENSE?**
> 10 mg/L held against a clearance of 11.55 L/hr needs 10 × 11.55 mg each hour, 115.5 mg/hr.

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a

Her key

Answer: 1000 mg

WHAT THE STEM GIVES

window	4 to 15 mg/L	*not needed: the target 10 mg/L is stated*
C_{ss}	10 mg/L	times Cl_T gives R
duration	12 hours	*not needed*
t½	6 hours	$k = \frac{0.693}{6} = 0.1155\ \text{hr}^{-1}$, the denominator
Cl_R	5	in L/hr, part of Cl_T
CIM	6.55 L/hr	$Cl_T = 5 + 6.55 = 11.55$ L/hr

weight	65 kg	*not needed*

THE WORKING, ONE LINE AT A TIME

ALGEBRA

$R = (10\ \text{mg/L})(11.55\ \text{L/hr}) = 115.5\ \text{mg/hr}$, with $Cl_T = 5 + 6.55 = 11.55\ \text{L/hr}$

Total clearance is the sum of the renal and metabolic clearances, and the rate that holds 10 mg/L is the target times that clearance.

SETUP

$$D_L = \frac{R}{k}, \text{ with } k = \frac{0.693}{6\ \text{hr}} = 0.1155\ \text{hr}^{-1}$$

A loading dose puts the steady-state amount in the body at once, and that amount is the rate over the elimination rate constant.

ALGEBRA

$$D_L = \frac{115.5\ \text{mg/hr}}{0.1155\ \text{hr}^{-1}} = 1000\ \text{mg}$$

mg/hr over hr^{-1}: the hours cancel and mg is left. It equals $C_{ss} \times V_D$, (10 mg/L)(100 L).

> **DOES THE NUMBER MAKE SENSE?**
> The volume is Cl_T over k, 11.55 over 0.1155, 100 L, and 10 mg/L in 100 L is 1000 mg.

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b

Her key

Answer: 720 mL

WHAT THE STEM GIVES

duration	12 hours	times 60 minutes, the infusion time
rate	1 mL/min	the volume delivered each minute
C_{ss}	10 mg/L	*not needed for the volume*
Cl_R	5	*not needed for the volume*
CIM	6.55 L/hr	*not needed for the volume*
window	4 to 15 mg/L	*not needed*
t½	6 hours	*not needed for the volume*
weight	65 kg	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

Volume = rate × time, over the 12 hours the concentration is maintained

A solution running in at 1 mL each minute for 12 hours delivers the rate times the time.

UNIT

$$\text{Volume} = (12\ \text{hr})\left(\frac{60\ \text{min}}{1\ \text{hr}}\right)(1\ \text{mL/min}) = 720\ \text{mL}$$

Hours to minutes is × 60, so the minutes cancel against mL/min and mL is left.

> **DOES THE NUMBER MAKE SENSE?**

12 hours is 720 minutes, and at 1 mL each minute the solution volume is 720 mL.

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b Her key

Answer: 1.925 mg/mL

WHAT THE STEM GIVES

duration	12 hours	the infusion time
rate	1 mL/min	gives the 720 mL volume
C_{ss}	10 mg/L	times Cl_T gives the drug rate
Cl_R	5	in L/hr, part of Cl_T
CIM	6.55 L/hr	$Cl_T = 5 + 6.55 = 11.55$ L/hr
window	4 to 15 mg/L	*not needed*
t½	6 hours	*not needed for the concentration*
weight	65 kg	*not needed*

THE WORKING, ONE LINE AT A TIME

ALGEBRA

$$R = (10 \text{ mg/L})(11.55 \text{ L/hr}) = 115.5 \text{ mg/hr}, \text{ with } Cl_T = 5 + 6.55 = 11.55 \text{ L/hr}$$

The drug rate that holds 10 mg/L is the target times total clearance; the litres cancel and mg/hr is left.

ALGEBRA

$$\text{Volume} = (12 \text{ hr})\left(\frac{60 \text{ min}}{1 \text{ hr}}\right)(1 \text{ mL/min}) = 720 \text{ mL}$$

The solution volume delivered at 1 mL/min over the 12 hours.

ALGEBRA

$$\text{Concentration} = \frac{(115.5 \text{ mg/hr})(12 \text{ hr})}{720 \text{ mL}} = \frac{1386 \text{ mg}}{720 \text{ mL}} = 1.925 \text{ mg/mL}$$

mg/hr × hr leaves the 1386 mg given in 12 hours; dividing by the 720 mL it is dissolved in gives mg/mL.

DOES THE NUMBER MAKE SENSE?

The solution runs in at 60 mL each hour and must contain the 115.5 mg given each hour, so about 1.9 mg in each mL; 1.925 mg/mL fits.

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c Her key

Answer: 100 L

WHAT THE STEM GIVES

C_{ss}	10 mg/L	*not needed for V_D*
t½	6 hours	$k = \frac{0.693}{6} = 0.1155 \text{ hr}^{-1}$
Cl_R	5	in L/hr, part of Cl_T
CIM	6.55 L/hr	$Cl_T = 5 + 6.55 = 11.55$ L/hr

weight	65 kg	*not needed: V_D comes from Cl_T and k*
window	4 to 15 mg/L	*not needed*
duration	12 hours	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

$$Cl_T = k \times V_D, \text{ so } V_D = \frac{Cl_T}{k}$$

Clearance is the rate constant times the volume of distribution; dividing clearance by k gives the volume.

ALGEBRA

$$V_D = \frac{11.55 \text{ L/hr}}{0.1155 \text{ hr}^{-1}} = 100 \text{ L}$$

L/hr over hr^{-1}: the hours cancel and litres are left. $Cl_T = 5 + 6.55 = 11.55$ L/hr and $k = \frac{0.693}{6 \text{ hr}} = 0.1155 \text{ hr}^{-1}$.

> **DOES THE NUMBER MAKE SENSE?**
> $V_D \times k$ must give back the clearance: 100 L × 0.1155 hr^{-1} = 11.55 L/hr, the sum of 5 and 6.55 L/hr.

d

Her key

Answer: 6 hr

WHAT THE STEM GIVES

C_{ss}	10 mg/L	the plateau; 5 mg/L is half of it
target	5 mg/L	50% of C_{ss}
t½	6 hours	$k = \frac{0.693}{6} = 0.1155 \text{ hr}^{-1}$
Cl_R	5	*not needed for the time*
CIM	6.55 L/hr	*not needed for the time*
window	4 to 15 mg/L	*not needed*
duration	12 hours	*not needed*
weight	65 kg	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_p = C_{ss}(1 - e^{-kt}): 5 \text{ mg/L} = (10 \text{ mg/L})(1 - e^{-(0.1155)t})$$

Without a loading dose the level rises from zero toward C_{ss} = 10 mg/L; 5 mg/L is 50% of C_{ss}.

ALGEBRA

$$0.5 = 1 - e^{-(0.1155)t}, \text{ so } e^{-(0.1155)t} = 0.5$$

Dividing both sides by 10 mg/L cancels the units and leaves the fraction of C_{ss} reached.

ALGEBRA

$$t = \frac{\ln 2}{0.1155\ \text{hr}^{-1}} = 6\ \text{hr}$$

A pure number over hr^{-1} gives hours, with $k = \frac{0.693}{6\ \text{hr}} = 0.1155\ \text{hr}^{-1}$: half of C_{ss} is reached in one half-life, 6 hours.

> **DOES THE NUMBER MAKE SENSE?**
> Half of C_{ss} is reached after one half-life whatever the rate, so 5 mg/L, half of 10 mg/L, takes 6 hours.

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e

Her key

Answer: 13.93 hr

WHAT THE STEM GIVES

t½	6 hours	$k = \frac{0.693}{6} = 0.1155\ \text{hr}^{-1}$
fraction	80%	$f = 0.8$, so $e^{-kt} = 0.2$
C_{ss}	10 mg/L	*not needed: only the fraction matters*
window	4 to 15 mg/L	*not needed*
duration	12 hours	*not needed*
Cl_R	5	*not needed: only k matters*
CIM	6.55 L/hr	*not needed*
weight	65 kg	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

$$0.8 = 1 - e^{-kt}, \text{ with } k = \frac{0.693}{6\ \text{hr}} = 0.1155\ \text{hr}^{-1}$$

Without a loading dose the fraction of C_{ss} reached at time t is $1 - e^{-kt}$.

ALGEBRA

$$e^{-(0.1155)t} = 0.2, \text{ so } t = \frac{-\ln 0.2}{0.1155\ \text{hr}^{-1}} = \frac{1.609}{0.1155\ \text{hr}^{-1}} = 13.93\ \text{hr}$$

The log of a pure number over hr^{-1} gives hours: just over two half-lives, since 75% takes two.

> **DOES THE NUMBER MAKE SENSE?**
> 75% of C_{ss} takes two half-lives, 12 hours, and 87.5% takes three, 18 hours; 80% lies between, a little after 12 hours, and 13.93 hr fits.

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f

Her key

Answer: 8.91 mg/L

WHAT THE STEM GIVES

C_{ss}	10 mg/L	the level when the infusion stops

t½	6 hours	$k = \frac{0.693}{6} = 0.1155\ hr^{-1}$
t	1, 6 and 12 hours	after the cessation: the exponent kt
window	4 to 15 mg/L	*not needed for the number*
Cl_R	5	*not needed: C_{ss} is given*
CIM	6.55 L/hr	*not needed*
weight	65 kg	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_p = C_{ss}\, e^{-kt}, \text{ with } k = \frac{0.693}{6\ hr} = 0.1155\ hr^{-1}$$

After the infusion stops the level falls first order from the steady-state 10 mg/L; t is counted from the stop.

ALGEBRA

$$C_1 = (10\ mg/L)\, e^{-(0.1155)(1)} = 8.91\ mg/L$$

kt = 0.1155 has no units, so the answer keeps mg/L; $e^{-0.1155} = 0.891$ of C_{ss} remains after 1 hour.

> **DOES THE NUMBER MAKE SENSE?**
> One hour is a sixth of a half-life, so a little over a tenth of the 10 mg/L is lost: 8.91 mg/L fits.

f

Her key

Answer: 5 mg/L

WHAT THE STEM GIVES

C_{ss}	10 mg/L	the level when the infusion stops
t½	6 hours	$k = \frac{0.693}{6} = 0.1155\ hr^{-1}$
t	1, 6 and 12 hours	after the cessation: the exponent kt
window	4 to 15 mg/L	*not needed for the number*
Cl_R	5	*not needed: C_{ss} is given*
CIM	6.55 L/hr	*not needed*
weight	65 kg	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_p = C_{ss}\, e^{-kt}, \text{ with } k = \frac{0.693}{6\ hr} = 0.1155\ hr^{-1}$$

After the infusion stops the level falls first order from the steady-state 10 mg/L; t is counted from the stop.

ALGEBRA

$$C6 = (10\ mg/L)\ e^{-(0.1155)(6)} = 5\ mg/L$$

kt = 0.693 has no units: 6 hours is one half-life, so half of the 10 mg/L remains.

> **DOES THE NUMBER MAKE SENSE?**
> Six hours is one half-life, so half of 10 mg/L, 5 mg/L, remains.

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f

Her key

Answer: 2.5 mg/L

WHAT THE STEM GIVES

C_{ss}	10 mg/L	the level when the infusion stops
t½	6 hours	$k = \frac{0.693}{6} = 0.1155\ hr^{-1}$
t	1, 6 and 12 hours	after the cessation: the exponent kt
window	4 to 15 mg/L	*not needed for the number*
Cl_R	5	*not needed: C_{ss} is given*
ClM	6.55 L/hr	*not needed*
weight	65 kg	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_p = C_{ss}\ e^{-kt},\ \text{with}\ k = \frac{0.693}{6\ hr} = 0.1155\ hr^{-1}$$

After the infusion stops the level falls first order from the steady-state 10 mg/L; t is counted from the stop.

ALGEBRA

$$C12 = (10\ mg/L)\ e^{-(0.1155)(12)} = 2.5\ mg/L$$

kt = 1.386 has no units: two half-lives leave a quarter of the 10 mg/L.

> **DOES THE NUMBER MAKE SENSE?**
> Twelve hours is two half-lives, so a quarter of 10 mg/L remains: 2.5 mg/L.

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g

Her key

Answer: 85.5 mg/hr

WHAT THE STEM GIVES

Cl_R new	2 L/hr	replaces the 5 L/hr
ClM	6.55 L/hr	unchanged: $Cl_T = 2 + 6.55 = 8.55$ L/hr
C_{ss}	10 mg/L	the same target
Cl_R	5	*not needed: replaced*
t½	6 hours	*not needed for the rate*
window	4 to 15 mg/L	*not needed*

duration	12 hours	*not needed*
weight	65 kg	*not needed*

THE WORKING, ONE LINE AT A TIME

ALGEBRA

$Cl_T = (2 + 6.55)$ L/hr = 8.55 L/hr

Total clearance is renal plus metabolic; only the renal part falls, to 2 L/hr, and the metabolic 6.55 L/hr is unchanged.

ALGEBRA

R = (10 mg/L)(8.55 L/hr) = 85.5 mg/hr

$R = C_{ss} \times Cl_T$: mg/L × L/hr leaves mg/hr. The lower clearance needs a lower rate to hold the same 10 mg/L.

> **DOES THE NUMBER MAKE SENSE?**
> The rate is the target times the new clearance: 10 × 8.55 = 85.5 mg/hr, lower than before because less drug is cleared each hour.

g

Her key

Answer: 1000 mg

WHAT THE STEM GIVES

Cl_R new	2 L/hr	replaces 5 L/hr: $Cl_T = 2 + 6.55 = 8.55$ L/hr
CIM	6.55 L/hr	unchanged
V_D	100 L	unchanged
C_{ss}	10 mg/L	the same target
t½	6 hours	*not needed: the old half-life no longer applies*
window	4 to 15 mg/L	*not needed*
duration	12 hours	*not needed*
Cl_R old	5	*not needed: replaced by 2 L/hr*
weight	65 kg	*not needed*

THE WORKING, ONE LINE AT A TIME

ALGEBRA

$Cl_T = (2 + 6.55)$ L/hr = 8.55 L/hr; $k = \frac{Cl_T}{V_D} = \frac{8.55 \text{ L/hr}}{100 \text{ L}} = 0.0855 \text{ hr}^{-1}$

The volume is unchanged at 100 L, so the lower clearance means a smaller k: the litres cancel and hr^{-1} is left.

ALGEBRA

R = (10 mg/L)(8.55 L/hr) = 85.5 mg/hr

The rate that holds 10 mg/L with the new clearance: mg/L × L/hr leaves mg/hr.

ALGEBRA

$$D_L = \frac{R}{k} = \frac{85.5 \text{ mg/hr}}{0.0855 \text{ hr}^{-1}} = 1000 \text{ mg}$$

mg/hr over hr^{-1} leaves mg. It equals $C_{ss} \times V_D$, (10 mg/L)(100 L): the loading dose depends on the volume, which has not changed.

DOES THE NUMBER MAKE SENSE?

The loading dose is 10 mg/L times the unchanged 100 L, so it stays at 1000 mg, the same as before the renal failure; only the rate falls.